

IIT Foundation Module for Class VI

Physics

- Theme 1: Physical Quantities and Measurement
 1. Measurement of Volume (3D concept)
 2. Measurement of Area
 3. Measurement of Density of Regular Solids
 4. Calculation of Speed

- Theme 2: Force and Pressure: Motion
 1. Motion as a change in position of an object with respect to time
 2. Types of motion
 3. Uniform and Non-Uniform Motion: concept of distance and speed (average speed)
 4. Weight

- Theme 3: Energy
 1. Energy
 2. Law of Conservation of energy

- Theme 4: Light Energy
 1. Reflection
 2. Laws of Reflection
 3. Plane mirror
 4. Speed of light (3×10^8 m/s)
 5. Primary colours (RGB)
 6. Formation of secondary colours by colour addition
 7. Appearance of colour of an object (based on reflection and absorption)
 8. Colour subtraction

- Theme 5: Heat

1. Heat as a form of energy and its units: Joule (J) and calorie (cal)
2. Different units of temperature ($^{\circ}\text{C}$, $^{\circ}\text{F}$, K)
3. Effects of heat
4. Methods of heat transfer
5. Thermos flask (application of heat transfer)

- Theme 6: Sound

1. Sources of sound
2. Sound as a longitudinal wave
3. Characteristics of a sound wave: Amplitude (relate amplitude with loudness) and Frequency
4. Sound needs a medium to propagate
5. Reflection and absorption of sound
6. Relative speed of sound in different mediums

- Theme 7: Electricity and Magnetism

1. Law of magnetism
2. Test for a magnet (by repulsion)
3. Electromagnetism, electromagnets and their applications – electric bell
4. Electric current as a flow of charge
5. Electric cell as a source of electricity
6. Resistors as components that oppose the flow of current
7. Symbolic representation of electrical components (key, battery, bulb, conducting wire, resistor)
8. Simple electric circuit – series and parallel
9. Battery as a collection of cells connected in series
10. Good and bad conductors of electricity